ViewGroups and Views in Android

1. ViewGroups

ViewGroups are containers that organize multiple Views and control their layout and positioning. They act as parents to Views and other ViewGroups.

Examples of View Groups

List of ViewGroups in Android

1. **LinearLayout** – Arranges Views in a single row or column.
2. **RelativeLayout** – Positions Views relative to each other or the parent.
3. **ConstraintLayout** – Flexible layout with constraints for precise positioning.
4. **FrameLayout** – A simple layout for stacking or overlaying Views.
5. **TableLayout** – Organizes Views into rows and columns like a table.
6. **GridLayout** – A flexible grid-based layout for arranging Views in rows and columns.
7. **ScrollView** – Allows vertical scrolling of content.
8. **RecyclerView** – A more advanced and efficient scrolling list of Views.
9. **ViewPager** – A swipeable layout for switching between different pages of content.

In XML ViewGroups are represented in tags.

Common ViewGroup tags include **<LinearLayout>**, **<RelativeLayout>, <ConstraintLayout>**, etc.

Example Linear Layout Usage in XML

Notice that it has some properties, width, height, padding

```
1  <LinearLayout
2      xmlns:android="http://schemas.android.com/apk/res/android"
3      android:layout_width="match_parent"
4      android:layout_height="match_parent"
5      android:orientation="vertical"
6      android:padding="16dp">
7
8      <!--  Views Goes Inside Here  -->
9
10 </LinearLayout>
```

2. Views in Android

Views are UI components inside ViewGroups that allow user interaction and display content.

Common Views Examples:

View	Description
TextView	Displays static text.
Button	Clickable button for actions.
EditText	User input field (e.g., text, email, password).
CheckBox	Selectable box that can be checked or unchecked.
ImageView	Displays images.

In this example:

- <TextView> displays a welcome message.
- <Button> provides a clickable interaction.
- <EditText> allows the user to enter text.
- <CheckBox> lets the user select an option.

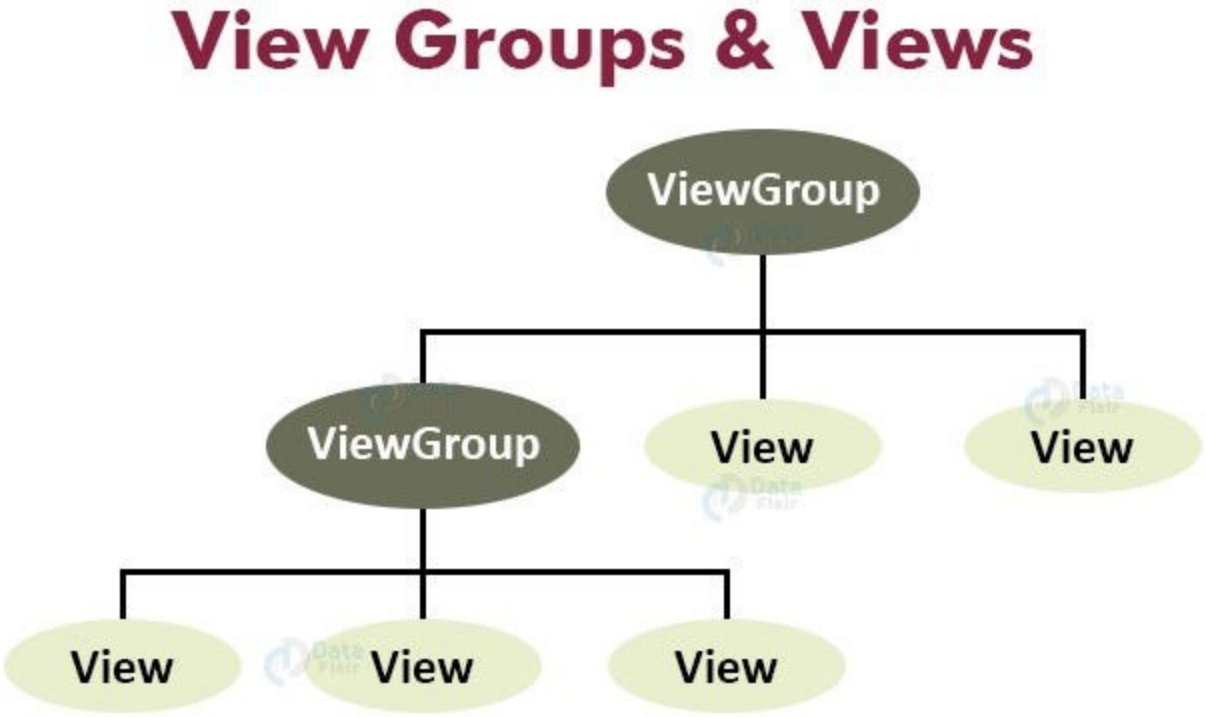
  Others include RadioButton, CardView, ProgressBar,

  Checkbox and Button usage in XML

```
<CheckBox
    android:text="Subscribe to Newsletter"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>

<Button
    android:text="Submit"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>
```

Below Diagram shows that a ViewGroup holds Views



Example of a **ViewGroup (Linear Layout)** and a Views **(Checkbox and Button)** arrangement.

```
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <CheckBox
        android:text="Subscribe to Newsletter"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <Button
        android:text="Submit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

Lets Look at Major Properties used in android ViewGroup and View

Common Properties of ViewGroups and Views

Since both ViewGroups and Views are XML elements, they share some **common properties** that define their appearance and behavior.

List of Common Properties:

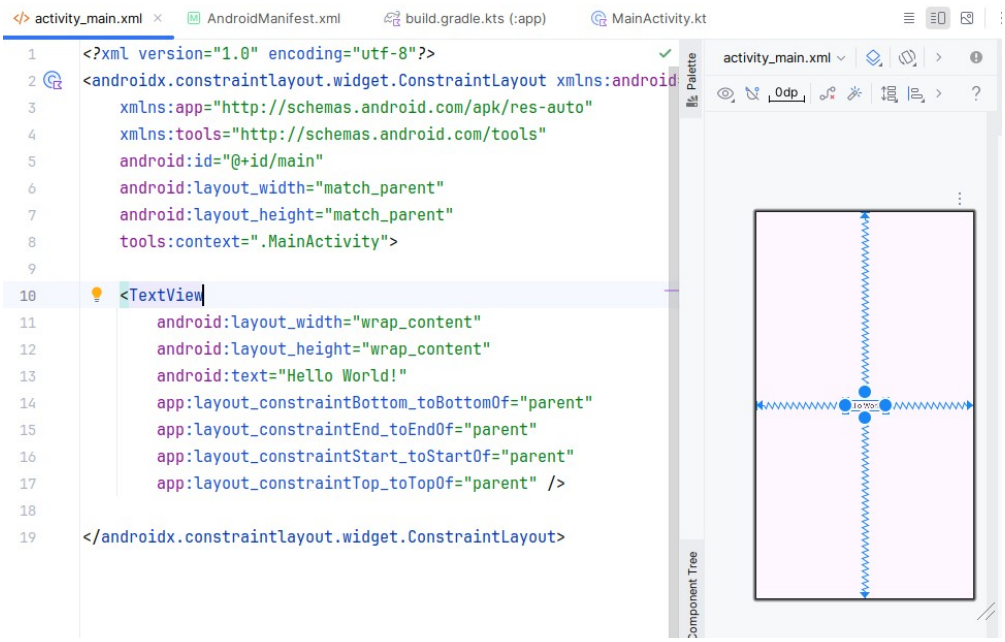
Property	Description	Example
android:layout_width	Defines the width of the View/ViewGroup.	android:layout_width="match_parent"
android:layout_height	Defines the height of the View/ViewGroup.	android:layout_height="wrap_content"
android:padding	Adds space inside the View, around its content.	android:padding="10dp"
android:margin	Adds space outside the View, separating it from others.	android:layout_margin="8dp"
android:background	Sets the background color or drawable of the View.	android:background="blue"

Next, we Build an application using Linear Layout (ViewGroup), and Views i.e TextViews, Buttons, EditText, CheckBoxes.

Open Android Studio and Open your Class Project.

In Android, The User Interface is Created using XML.
In App File Structure, Go to **res – layout – activity_main.xml**

Below is the current look of our **activity_main.xml**



Next we update our activity_main.xml with Below Layout.

```
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:layout_margin="5dp"
    android:background="#F5F5F5">

    <!-- TextView Section -->
    <TextView
        android:id="@+id/heading"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="WELLNESS-HUB"
        android:textColor="#FF9800"
        android:layout_marginTop="80dp"
        android:textSize="40sp"
        android:textStyle="bold"
        android:textAlignment="center"/>

    <!-- Buttons Section -->
    <Button
        android:id="@+id/diets"
        android:text="Diets"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:backgroundTint="#3F51B5"
        android:textColor="#FFFFFF"
        android:padding="12dp"
        android:layout_marginTop="20dp"
        android:marginBottom="10dp"/>

    <Button
        android:id="@+id/exercises"
        android:text="Exercises"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:backgroundTint="#3F51B5"
        android:textColor="#FFFFFF"
        android:padding="12dp"
        android:layout_marginTop="20dp"
        android:marginBottom="20dp"/>

    <Button
        android:id="@+id/tips"
        android:text="Healthy Tips"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:backgroundTint="#3F51B5"
        android:textColor="#FFFFFF"
        android:padding="12dp"
        android:layout_marginTop="20dp"
        android:marginBottom="10dp"/>
```

```
    <!-- Email Input -->
    <EditText
        android:id="@+id/email"
        android:hint="Enter your email"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:padding="12dp"
        android:layout_marginTop="30dp"
        android:textColor="#000000"
        android:textSize="16sp"
        android:inputType="textEmailAddress"
        android:marginBottom="10dp"/>

    <!-- Checkbox -->
    <CheckBox
        android:id="@+id/newsletter"
        android:text="Send me the newsletter"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="16sp"
        android:textColor="#000000"
        android:layout_marginTop="30dp"/>

</LinearLayout>
```

Run App in your phone
Output

